ZOO504 – Wildlife (Subjective)

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SULMAN ALI (BS-ZOOLOGY)

(0341-7547044)



Name 2 wildlife sanctuaries of KPK:

Khyber PakhtunKhwa(KPK): 1= Manglot, 2= Argam Basti

How animals help to maintain hygienic conditions of environment:

The animal help to maintain hygienic conditions by 1. Nutrients recycling 2. Humus formation 3. Cleaning of environment.

Note on Hub dam wildlife sanctuary

Hab Dam Wildlife Sanctuary was declared in July 1974. It is Pakistan's third largest dam.

Area: 27,219ha

Location: It is situated in the north of Karachi bordering the Kirthar National Park Balochistan.

The greater part of this wildlife sanctuary in Balochistan is unprotected; the eastern shore and area south of the Dam in Sindh are protected in the Kirthar National Park and Hub Dam Wildlife Sanctuary respectively. The lake formed by the dam covers an area of 32 square miles and is an ideal place for bird watching. There is plenty of waterfowl in the lake, both resident and migratory.

Name the mammals of Hazarganji Chiltan:

Indian wolf • Stripped hyena • Leopard • Caracal • Jackal • Red fox • Porcupine

What are restrictions on wild life trade:

Restrictions on Import and Export of Animals, Trophies or Meat No person shall.

- Receive by gift, purchase or otherwise any animal of a kind specified in the Second Schedule.

- ❖ Unless a valid certificate of lawful possession in respect thereof.
- ❖ Restrictions on Dealing in Animals, Trophies and Meat.
- ❖ Wildlife Sanctuary.
- ❖ National Park

Game Reserve

Games reserves according to provinces name.

"A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis"

A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport.

There are 96 99 Game reserves in Pakistan ➢ Federal territory 1 ➢ Balochistan 7 ➢ Northern Area 9 ➢ Sindh 14 ➢ Punjab 19 ➢ KPK 38 ➢ AJK 8

1= In the game reserves the major focus is specifically the animals. 2= If hunting is prohibited, a game reserve may be considered a "Nature Reserve" 3= Wherein all aspects of naturally-occurring life in the area are considered. 4= Most of the areas in game reserves have created to provide habitat protection for animal species commonly referred to as game (huntable species for sport or meat)

Importance of scientific wildlife/ or significance of wild life:

Many wildlife studies focus on conservation and management, with the aim of learning about the ecology of a population in the field. In such cases, minimising disturbance to the animals is important for the scientific validity of the research as well as for good animal welfare. Wild life provides a stability to different natural processes of nature. The importance of wildlife can be categorized as ecological importance, economic importance, investigatory importance, conservation of biological diversities etc. Wildlife helps in maintaining the balance of nature. One of the most vital roles of wildlife conservation to humans is food security. Through the protection of forests against deforestation as well as restoring natural habitats, economic opportunities would rise. Moreover, wildlife conservation helps in promoting agricultural diversity.

Ramser sites of Sindh?

Sindh • Keenjhar (Kalri) Lake • Haleji Lake • Drigh Lake • Indus Dolphin reserve • Jubho lagoon • Nurri Lagoon • DehAkro-II • Indus Delta • Runn of Kutch.

State Purpose of wildlife sanctuaries.

Animal sanctuaries exist for many specific reasons: 1. Help to protect animals and safeguard their lives. 2. Animal sanctuaries give new homes to abandoned animals. 3. Animals rescued from dangerous conditions. 4. Help to protect animals from illegal activities. 5. Serve as places where endangered animals breed and are therefore protected.

What is Poaching?

Poaching is the

1= Illegal killing for tusks, oil, decoration (as stuffed preparation), Plumage. 2= Illegal hunting for bush meat, trophies.

Effects of pollution:

Pollution of various kinds have adversely affected wild animals.

Insecticides: There has been a considerable damage to birds by insecticides spraying.

Oil Spillage: Oceanic birds are killed by oil spillage from oil tankers in the sea.

Noise Pollution: It causes disturbance of hormonal and reproductive behavior of many wild animals.

Water Pollution: Domestic as well as industrial waste dumped into water bodies causes decline in stocks of fishes.

- ❖ Animals like mussels have been unfit for human consumption
- ❖ Various natural calamities like floods, hurricane, fire etc. have also caused massive destruction of wildlife.

Categories of IUCN:

1. Category I
- ❖ Category Ia
- ❖ (Strict Nature Reserve)
- ❖ Category Ib (Wilderness Area)
2. Category II (National Park)
3. Category III (Natural Monument or Feature)
4. Category IV (Habitat/Species Management Area)
5. Category V (Protected Landscape/\Seascape)
6. Category VI (Protected area with sustainable use of natural resources)

HOW dispersal seeds occur?

Dispersal of seeds------ (squirrels, birds, rodents etc.) Seeds can stick to fur . Squirrels or birds pick them up and drop or bury them. Seed eaters also disperse the seeds by defecation . Caching behavior of many rodents and some birds.

Acts prohibited under law of national park:

Acts prohibited under Law in National Parks:

• Firing of guns • Hunting, killing, trapping or capturing wild animals inside or within a radius of three miles of its boundaries. • Interfering with animals and plants • Clearing land for cultivation. • Flow of polluted water is also prohibited.

Major Challenge of vaterniarans:

- ❖ Environmental contaminants
- ❖ Infectious diseases
- ❖ Outbreaks that potentially threaten wild and domesticated animals as well as humans. Wildlife veterinarians are necessary and should made available to care of wild animals.

Veterinarians are to prevent

- ❖ Infectious disease transfer at the domestic animal-wildlife interface.
- ❖ Movement of diseases between wildlife and domestic species.
- ❖ Reducing contact between domestic and wild animals where necessary

What is pollution:

Pollution is anything that makes the earth dirty and unhealthy. Land, air, and water are all affected by pollution. Pollution takes up space on our land. Many of the things people use every day come in packages, like food, games, school supplies, and electronics.

What is monitoring trade in CITES.

Convention on International Trade in Endangered Species (C.I.T.E.S)

Monitoring trade:

- ❖ The backbone of CITES is the permit system that facilitates international cooperation in conservation and trade monitoring.
- ❖ Permits are issued only if a country's Management and Scientific Authorities determine that trade is legal and does not threaten the species' survival in the wild.
- ❖ These permit allows inspection at ports of export and import to quickly verify that CITES specimens are properly documented.

- ❖ This trade monitoring has created an information on the management and use of CITES species worldwide.
- ❖ CITES has helped global conservation of species.
- ❖ CITES provides tools to effectively conserve the world's diverse natural resources.
- ❖ It control global over-exploitation of wildlife.

Three safety rules to save game reserves.

Driving Areas: Vehicles must remain on the designated roads at all times.

Walking Areas: Walking trails must be provided throughout the reserve.

Feeding of Wildlife is Prohibited: The feeding or intentional disturbance of wildlife is a serious offence.

What is territory?

Territory is the sociographical area that an animal of a particular species consistently defends against conspecifics (or, occasionally, animals of other species). Animals that defend territories in this way are referred to as territorial. Territoriality is only shown by a minority of species.

What is musk deer?

Musk deer can refer to any one, or all seven, of the species that make up Moschus, the only extant genus of the family Moschidae. The musk deer are lacking antlers.

Note on IUCN:

International Union for Conservation of Nature and Natural Resources (I.U.C.N)

IUCN was founded in 1948. It is leading international, non-government organization concerned with conservation. Network of governmental and non-governmental organization.

Members: IUCN has more than: 475 member organizations. 116 government agencies in 133 countries. Functions: Monitoring the status of conservation. Developing plans for dealing with problems such as the World Conservation strategy. Promoting action on their plans by government or organizations. Coordinating communication between the members and the commissions as well as the development. Providing advice and assistance to complement conservation measures. Selection and management of WWF projects. Managing the UNEP and C.I.T.E.S secretariat. Performing the continuing duties under conservation of wetlands of international importance, specially by waterfowl habitats.

What is the IRDCP project.

Introduction The Indus River Dolphin Conservation Project (IRDCP) was developed to conserve a viable population of the Indus River Dolphin (Platanista gangetica minor), an endangered species endemic to the Indus River system in Pakistan.It is one of the only four extant freshwater dolphin species in the world. The habitat of this species has been reduced to one fifth of its historical range. The remaining habitat is being further degraded primarily due to shortage of water caused by its diversion to meet growing agricultural needs. In addition, these diversions also reduce the flow of water in the river, thus increasing the incidence of pollutant accumulation. The population of the Indus River Dolphin is divided in sub-populations because of the six barrages constructed on the River Indus. Moreover, Canal strandings, contamination due to industrial waste and agrochemicals, unsustainable fishing and net entanglements also contribute to the species'endangerment.

Project Area: This project will cover the Indus Dolphin Reserve, covering an area of 200 km between Guddu and Sukkur barrages in the province of Sindh.

Project Objectives: 1= Protecting the innate biodiversity of the Lower Indus River Basin Eco-system. 2= Ensuring the sustainable use of riverine biodiversity.3= Promoting actions to mitigate pollution and the wasteful extraction of riverine resources.

Causes of depletion of wildlife:

The main factors of Persecution are:

Wildlife Trade: Trading of wildlife and its products. e.g. Bush meat, Ivory, hide, trophy, fur etc.

Trapping of Wild Animals:

- Pets(cats etc.)
- Sold for Medical experimental studies (Monkeys, Rabbits etc.)
- For street shows, circuses (Bears,Parrots,Lions)

Poaching

- Illegal killing for tusks, oil, decoration (as stuffed preparation), Plumage.
- Illegal hunting for bush meat, trophies.

Absurd killing: Tigers, jackals, wolves etc. are killed by the farmers.

Ramsar sites are wetlands of international importance designated under the Ramsar Convention. They can include marshes, lakes, rivers, and coastal areas. There are over 2,400 Ramsar sites worldwide,

What are the Ramser sites:

A Ramsar Site is a wetland site designated to be of international importance. These wetlands are protected under strict guidelines of the Ramsar Convention on Wetlands.

What are common Hydrological alternation in wetland area.

Common hydrologic alterations in wetland areas include: 1. Deposition of fill material for development. 2. Drainage for development, farming, and mosquito control. 3. Dredging and stream channelization for navigation, development, and flood control. 4. Diking and damming to form ponds and lakes. 5. Diversion of flow to or from wetlands. 6. Addition of impervious surfaces in the watershed, thereby increasing water and pollutant runoff into wetlands.

What is the tularemia?

Tularemia (rabbit fever)

Host: Rodents and rabbits. fox, beaver, mice, and muskrats and some species of birds.

Transmission: Handling infected animals, contamination of cuts, mosquito/fly/ tick bites, and by inhalation.

How climate change effects the environment:

“any change in climate over time, whether due to natural variability or as a result of human activity.” Climate change is also taking its toll in wetland degradation.

More frequent and intense drought, storms, heat waves, rising sea levels, melting glaciers and warming oceans can directly harm animals, destroy the places they live, and wreak havoc on people's livelihoods and communities. As climate change worsens, dangerous weather events are becoming more frequent or severe.

Musk deer is a fascinating creature found in various parts of Asia. They are small, deer-like animals known for their musk glands, which produce a strong-smelling substance called musk. Musk deer are known for their solitary nature and their unique adaptation of long, sharp canine teeth. Unfortunately,they are also facing threats due to habitat loss and illegal hunting for their musk gland

Territory refers to an area that an animal or a group of animals defends and considers its own. It's like their personal space that they protect from other individuals of the same species. Animals mark their territory through various means, such as scent marking, vocalizations, or physical displays. Having a territory helps animals establish their dominance, find resources like food and shelter, and attract mates. It's their way of maintaining their own little piece of the world!

In wetland areas, common hydrologic alterations can include changes in water levels, flow patterns, and overall water availability. These alterations can be caused by various factors such as human activities, climate change, or natural events like floods or droughts. They can have significant impacts on the wetland ecosystem, affecting the vegetation, wildlife, and overall ecological balance. It's important to monitor and manage these hydrologic changes to ensure the long-term health and sustainability of wetland areas.

Climate change has various effects on the environment. Rising temperatures can lead to the melting of glaciers and polar ice, causing sea levels to rise and coastal areas to flood. It can also disrupt weather patterns, leading to more frequent and intense extreme weather events like hurricanes, droughts, and heatwaves. Climate change can impact ecosystems, causing shifts in habitats and affecting the distribution of species. It can also lead to changes in rainfall patterns, affecting agriculture and water resources.